

Story 1

Sifting through the Snow: Cleaning and Exploratory Analysis

The Digital Sorting Hat: Machine Learning Segment Discovery

The
Oper
ation
al X..

The Story:

The Goal: We began with 1.1 million records of February 2022 Citi Bike data to identify how winter conditions affect urban mobility.

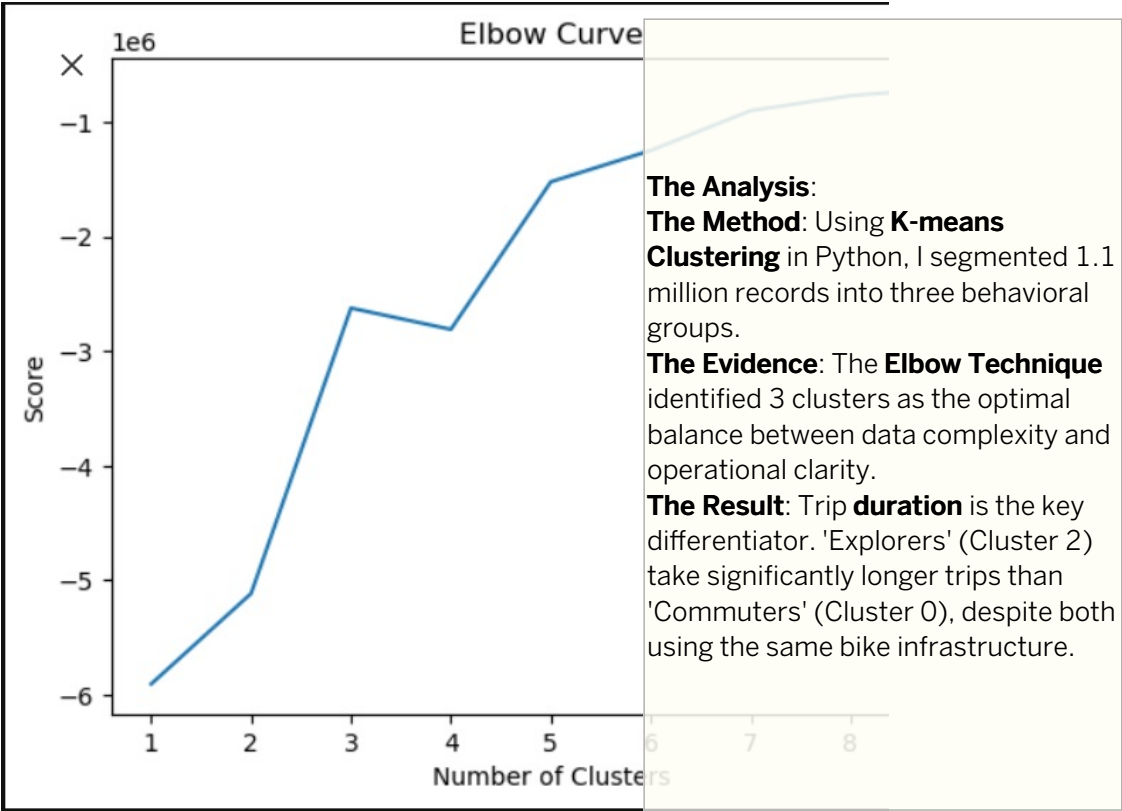
Early Questions: We asked: Is there a significant difference in how subscribers (Members) vs. one-time users (Casual) use the system during sub-zero temperatures?.

The Prep: Rigorous data cleaning was performed to handle mixed-type data and ensure consistency across coordinates and timestamps.

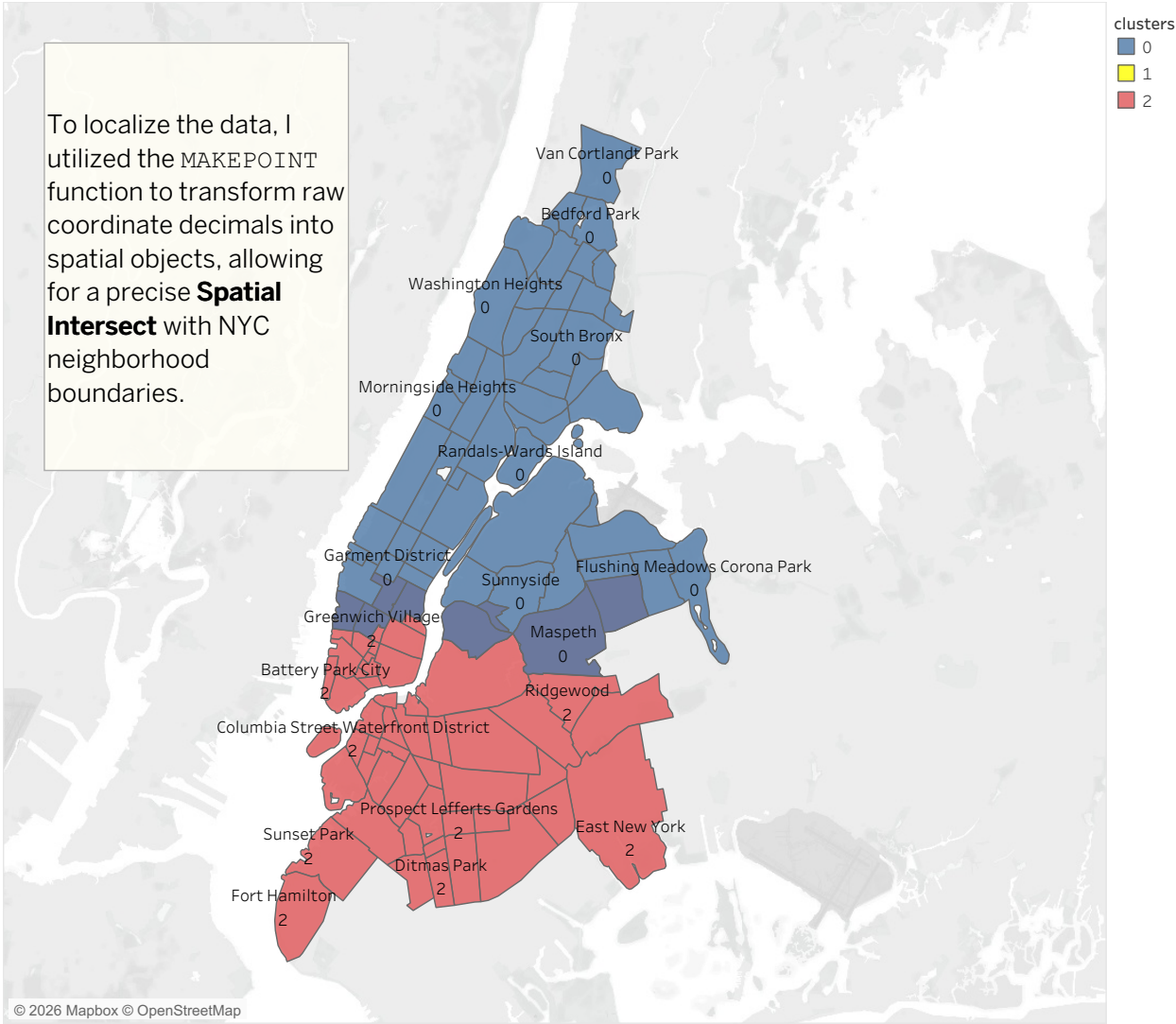
Initial Discovery: Our exploratory analysis showed that while trip volume is lower in winter, "Member" ridership remains highly consistent, suggesting a strong commuter base.

Story 1

Sifting through the Snow: Cleaning and Exploratory Analysis	The Digital Sorting Hat: Machine Learning Segment Discovery	The Operational X-Ray: Mapping Segment Hubs
---	---	---



Story 1



Story 1

The Operational X-Ray: Mapping Segment Hubs	Strategic Logistics & Maintenance Playbook	
---	--	--

1. Strategic Action: The "North-South" Rebalancing

The Insight: Cluster 0 (Manhattan) shows high-frequency, short-duration "Commuter" behavior, while Cluster 2 (Brooklyn/Queens) displays leisure-oriented "Explorer" patterns.

The Action: Prioritize bike rebalancing in Manhattan commercial hubs during Monday-Friday AM peaks to support the "Commuter" cluster.

The Action: Increase dock availability along the Brooklyn and Queens waterfronts on weekends to accommodate the "Explorer" cluster.

2. Maintenance Action: The "Cluster 1" Hardware Audit

The Insight: Cluster 1 represents "Zero-Distance" trips with abnormally high durations, suggesting users are struggling to undock bikes or find working docks.

The Action: Target the specific stations identified in the Cluster 1 hotspots for immediate hardware inspection and dock repair.

3. Project Limitations & Future Scope

Seasonality: This analysis represents **February 2022**; results may shift significantly in summer months as weather influences rider endurance.

Data Gaps: The current dataset lacks user demographic data (age, income), which limits our ability to build true "persona-based" marketing.

Next Step: Future iterations should integrate **National Weather Service (NOAA)** data to predict ridership spikes based on "RealFeel" temperatures.

Story 1

The Oper ation al X-..	Strategic Logistics & Maintenance Playbook	
---------------------------------	--	--

